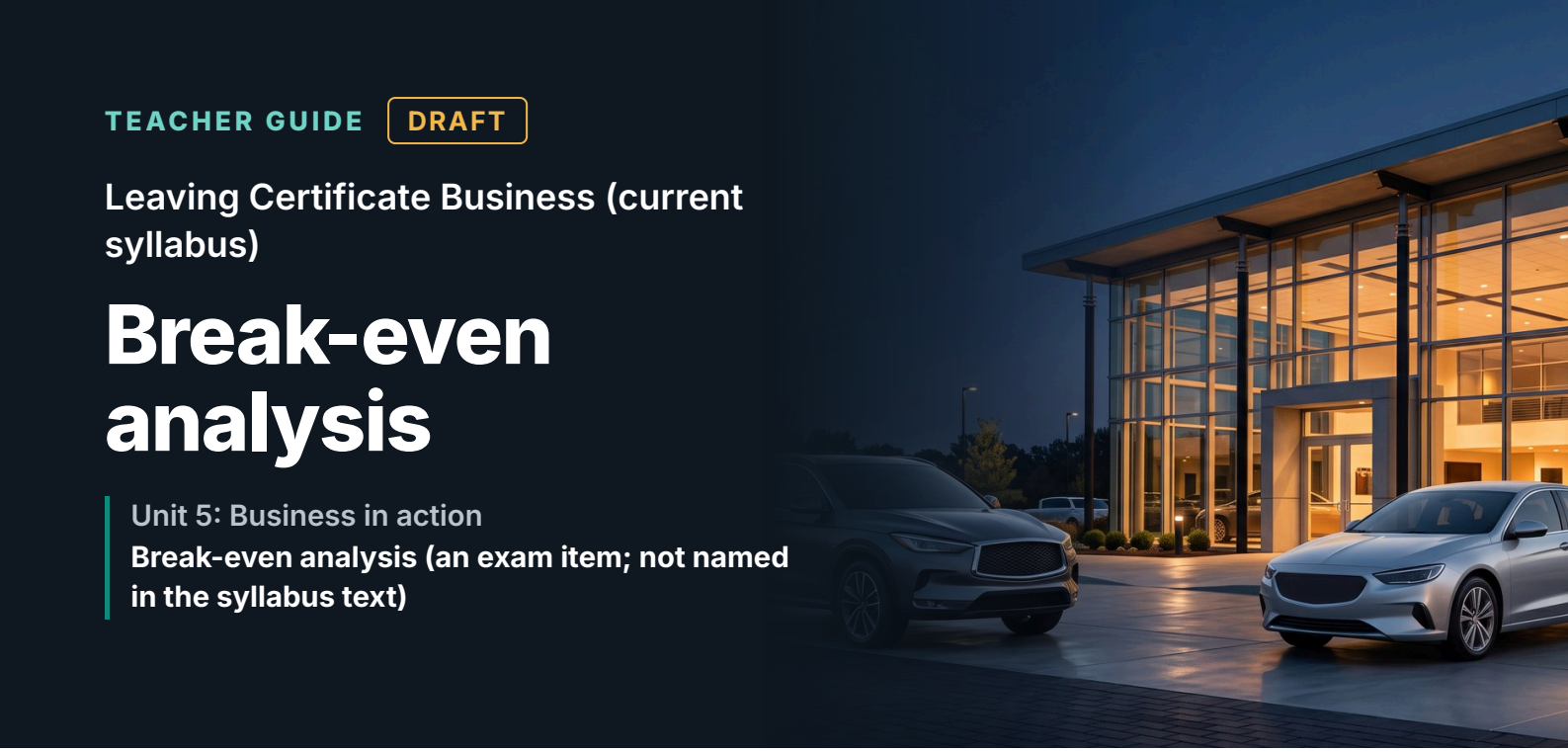
TEACHER GUIDE DRAFT

Leaving Certificate Business (current syllabus)

Break-even analysis



Unit 5: Business in action

Break-even analysis (an exam item; not named in the syllabus text)

A revision lesson. Pupils run a car dealership in firms of four and revise break-even analysis on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min	45 min	60 min	75 min

What pupils do

Every pupil buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal stock

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Cars to break even

Round 1

They take up to 5 cars.

Cars to break even? Round up.

Livery, one-off	**€2,400**
Staff training, one-off	**€2,150**
Customer pays per car	**€18,760**
You pay per car	**€16,800**

ANSWER

3 cars

(€2,400 + €2,150) ÷ €1,960 = 2.32 → 3 cars

Why: Part of a car does not sell: 3 whole cars cover the costs.

2 Break-even limits

Round 1

You plan a fleet deal with a break-even chart.

Which two are limits of break-even analysis?

- **A** It leaves out the fixed costs
- **B** It cannot show the profit
- **C** It assumes one fixed price
- **D** It assumes every car is sold
- **E** It needs the firm's cash balance

ANSWER

✓ It assumes one fixed price
✓ It assumes every car is sold

Why: It assumes all output sells at one price; it does show fixed costs and profit.

3 Safety margin as %

Round 2

Three orders, each with its break-even point.

Order by margin of safety as a %, lowest first?

- **A** Order 8 cars, break-even 6
- **B** Order 20 cars, break-even 16
- **C** Order 10 cars, break-even 7

ANSWER

1. **Order 20 cars, break-even 16**
2. **Order 8 cars, break-even 6**
3. **Order 10 cars, break-even 7**

Why: (20 − 16) ÷ 20 = 20%, (8 − 6) ÷ 8 = 25%, (10 − 7) ÷ 10 = 30%.

Topic tasks

4 Profit on the deal

Round 2

A deal with a set-up cost.

Profit on the deal?

Livery and set-up, one-off	**€6,550**
Customer pays per car	**€12,475**
You pay per car	**€10,900**
They order	**9 cars**

ANSWER

€7,625

9 × (€12,475 − €10,900) − €6,550 = €7,625

Why: The first cars pay the set-up; the rest is profit: €7,625.

5 Which price to set

Round 3

You expect 7 sales at either price. At €9,500 a car you break even at 6; at €9,000, at 8.

Which price should you set, and why?

A €9,000 a car

B €9,500 a car

ANSWER

✔ **€9,500 a car**

Why: At 7 sales €9,500 clears its break-even of 6; €9,000 falls short of 8.

6 Margin after rise

Round 3

They order 8 cars.

Margin of safety now, in cars? Round break-even up.

Livery and set-up, one-off	**€7,900**
Customer pays per car	**€16,665**
Cost per car before the rise	**€13,150**
Cost rise	**10%**

ANSWER

4 cars

€13,150 + 10% = €14,465; €7,900 ÷ €2,200 → 4; 8 − 4 = 4

Why: The cost rise ate into the margin of safety: 4 cars left.

Final questions

Each pupil answers 6 final questions. Pick the question topic “Break-even analysis”: the 6 come from these 10 questions.

1 Fixed costs are €24,000 a month. Each car sells for €9,000 and costs €7,000. How many cars to break even?
✔ **12 cars**

2 Contribution per valet is €40 and fixed costs rise from €18,000 to €20,000 a month. What is the new break-even?
✔ **500 valets**

3 What is contribution per unit?
✔ **Selling price minus variable cost per unit**

4 At break-even, what is true?
✔ **Total revenue equals total costs**

5 Price €1,200 per service package, variable cost €700, fixed costs €15,000. What is the break-even in packages?
✔ **30 packages**

6 Break-even is 40 cars and forecast sales are 55 cars. What is the margin of safety?
✔ **15 cars**

7 Variable cost per car rises because of a weaker euro. Nothing else changes. What happens to break-even output?
✔ **It rises**

8 Which cost is fixed for a dealership?
✔ **Showroom rent**

9 Fixed costs €30,000, price €500 and variable cost €300 per job. What is the profit at 200 jobs?
✔ **€10,000**

10 Break-even is 60 cars and each car contributes €1,500. What are the fixed costs?
✔ **€90,000**